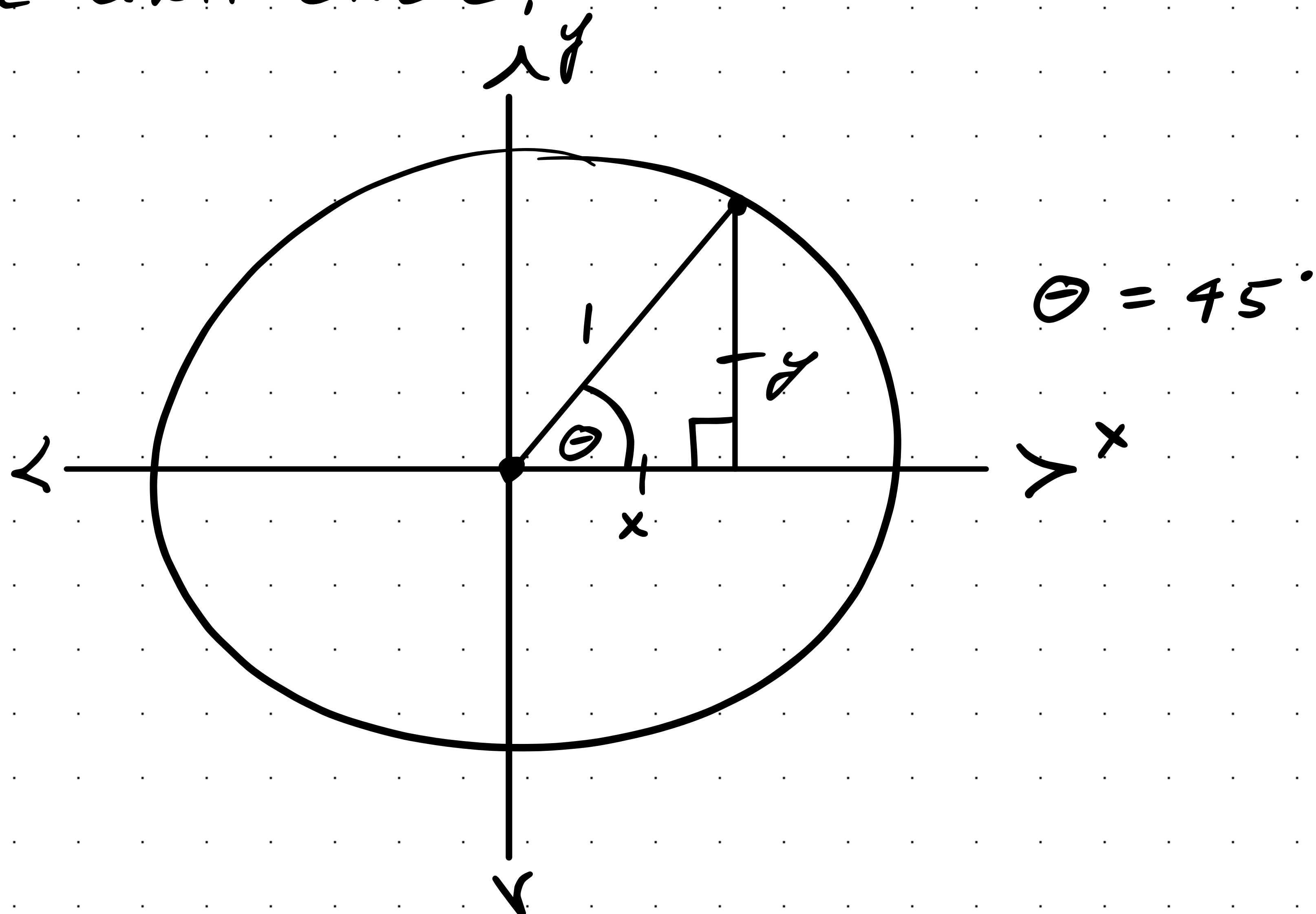


Given the unit circle,



45-45-90

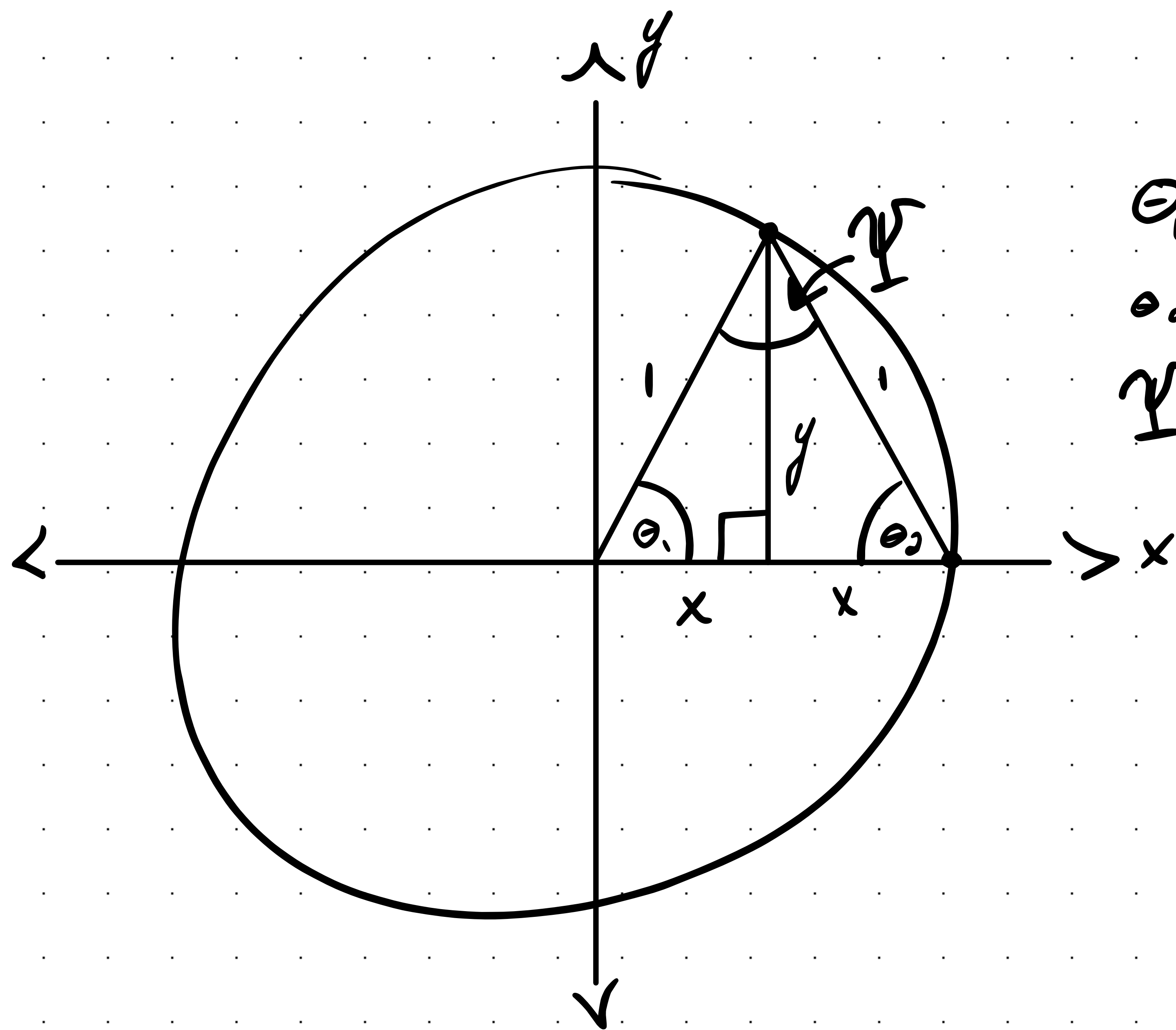
$$x = y$$

$$x^2 + y^2 = 1^2 \rightarrow 1 = x^2 + x^2 = 2x^2$$

$$x^2 = \frac{1}{2} \rightarrow x = \sqrt{\frac{1}{2}} = \frac{\sqrt{1}}{\sqrt{2}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$\therefore x = \frac{\sqrt{2}}{2}$$

$$y = \frac{\sqrt{2}}{2}$$



$$\theta_1 = 60^\circ$$

$$\theta_2 = 60^\circ$$

$$\psi = 60^\circ$$

30-60-90

$$\theta_1 = 60^\circ. \quad \frac{60^\circ}{2} = 30^\circ.$$

A 30° line intersects the reflected hypotenuse halfway along its length, which is 1.

Since the equilateral triangle composed of the original 30-60-90 and its reflection has three 60° angles, all three of its sides are equal in length, so the base is 1 unit wide, and half of this is $\frac{1}{2}$, therefore $x = \frac{1}{2}$ and $y = \frac{\sqrt{3}}{2}$

by Pythagorean theorem.

Q.E.D.